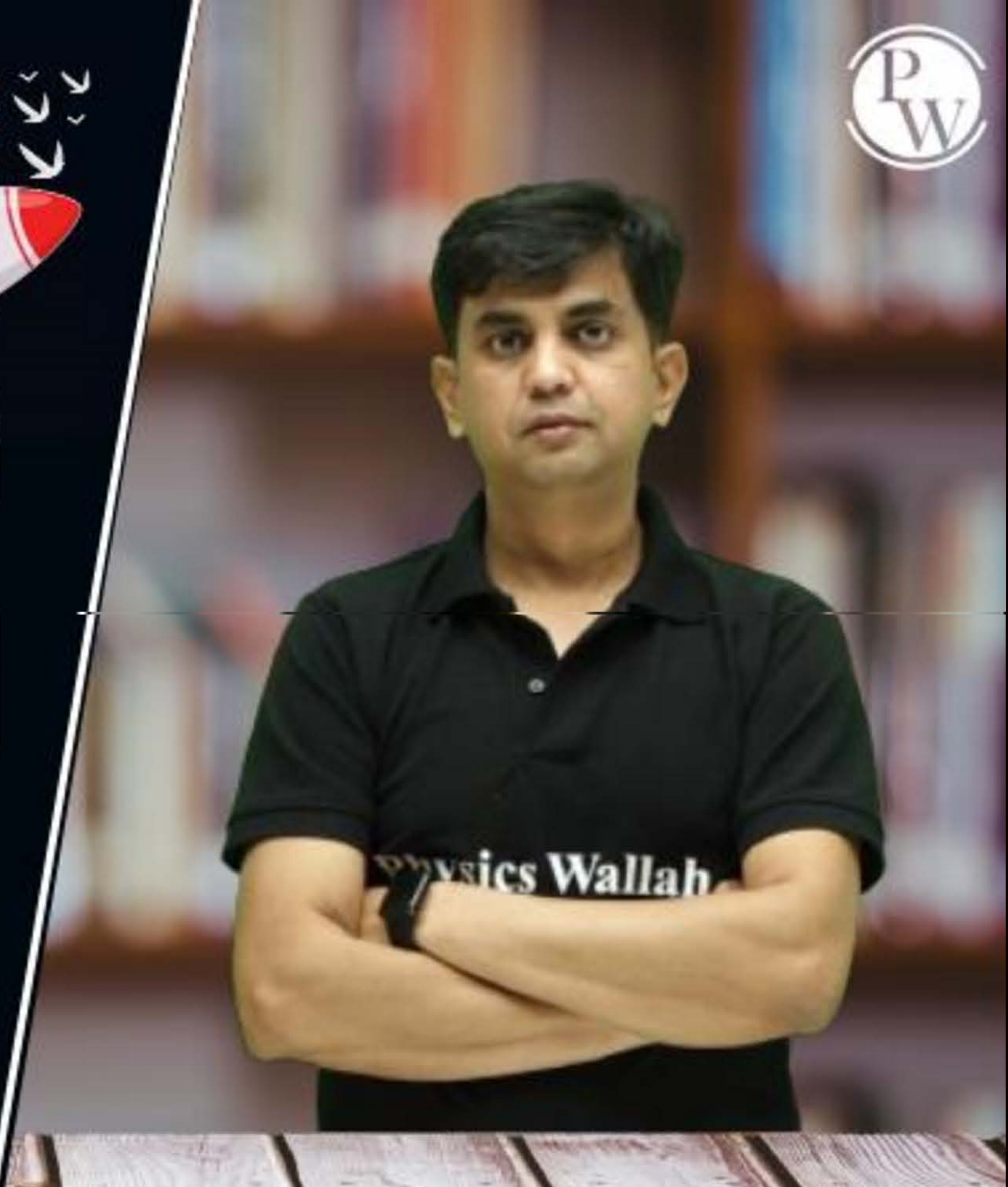
CS & IT ENGINEERING

Computer Network

Transport Layer

DPP - 01 Discussion Notes



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[MCQ]

#Q. UDP provides ________ services.

A 'Connection Less' and 'Unreliable'

B ~~'Connection Less' and 'Reliable'~~

C ~~'Connection Oriented' and 'Unreliable'~~

D ~~'Connection Oriented' and 'Reliable'~~

Ans: A

[NAT]

#Q. Consider total length field value of UDP packet header is 200 (in decimal), calculate its payload size (in bytes) ?

UDP Segment Size = Total Length = 200 bytes

UDP Header Size = 8 byte

UDP Segment Payload Size = (200 − 8) bytes

= 192 bytes

$\boxed{\text{Ans} = 192}$

[MCQ]

#Q. Which transport layer protocol is used for faster communication ?

A ~~TCP~~ = UDP + Extra services

B UDP → Basic Protocol

C ~~IP~~ → Network Protocol

D None of the above

$\boxed{\text{Ans: B}}$

[MCQ]

#Q. Which transport layer protocol is used by HTTP/1.1 (version 1.1) ?

A TCP

~~B~~ UDP

~~C~~ IP

D None of the above

$\boxed{\text{Ans: A}}$

[MSQ]

#Q. Identify $\boxed{\text{INCORRECT}}$ statement(s) regarding TCP and UDP.

A Flow Control provided by both TCP and UDP FALSE

B Flow Control provided by TCP, but not UDP TRUE

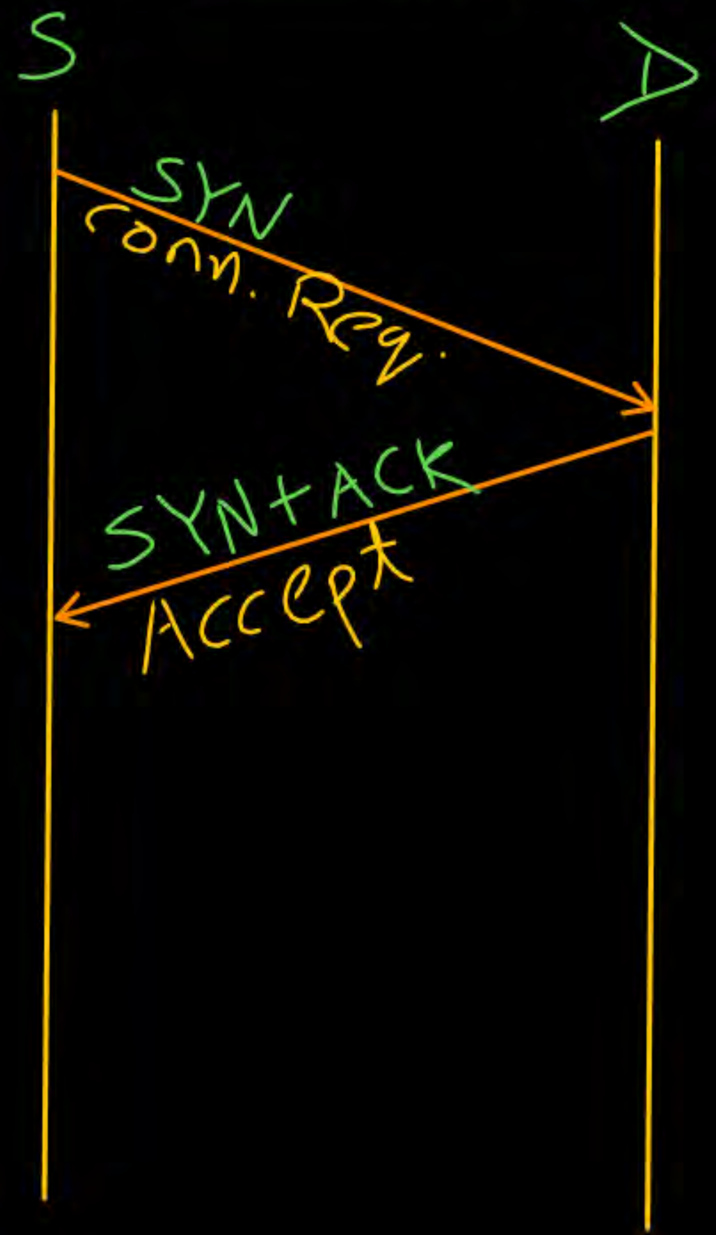
C Flow Control provided by UDP, but not TCP FALSE

D Flow Control provided by neither TCP nor UDP FALSE

$\boxed{\text{Ans: A, C \& D}}$

[MCQ]

#Q. Consider the 3-way handshake protocol for TCP connection establishment between hosts S and D. Host S sends a TCP connection request message (TCP segment) to D, and D accepts the connection request and send accept message (TCP segment) to S. Identify correct TCP header flag bits in accept message (TCP segment) ?



A. ~~SYN bit = 0 and ACK bit = 0~~

B. ~~SYN bit = 0 and ACK bit = 1~~

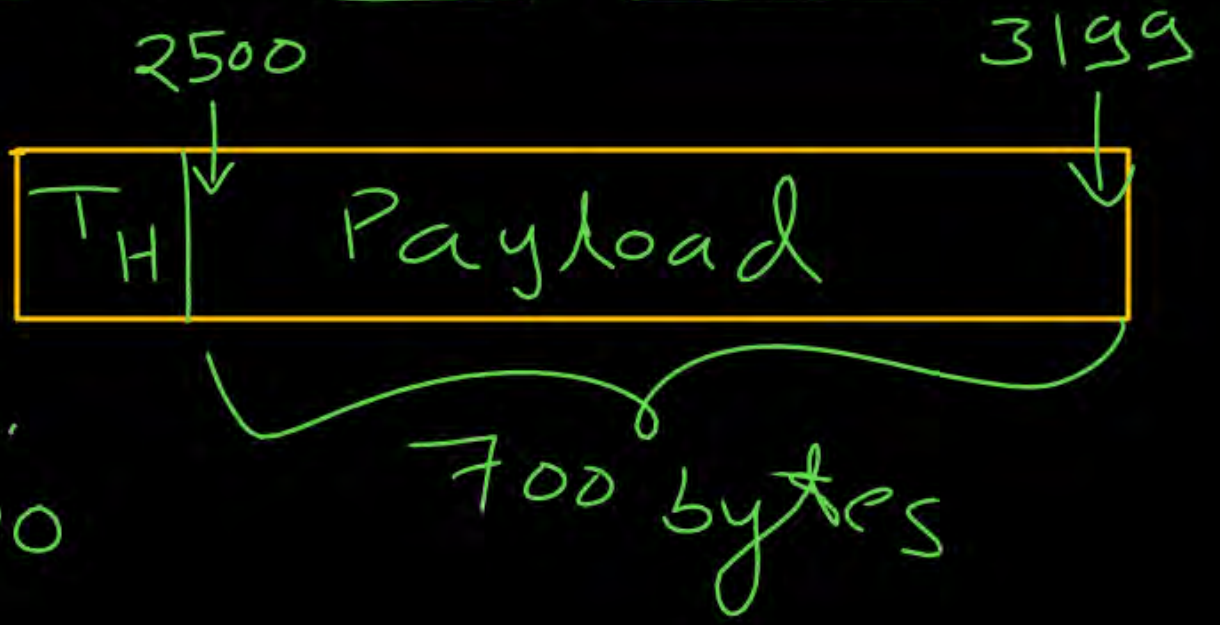
C. ~~SYN bit = 1 and ACK bit = 0~~

D. SYN bit = 1 and ACK bit = 1 ✓

$\boxed{\text{Ans: D}}$

[NAT]

#Q. Consider a TCP sender sends one segment in which sequence number field value (in decimal) is 2500 and the segment carried 700 bytes of data in its payload. What should be the acknowledgment number sent by receiver to acknowledge this segment successfully ?



Seq. No. = 2500

Ack No. = 2500 + 700
= 3200

$\boxed{\text{Ans: } 3200}$

[MCQ]

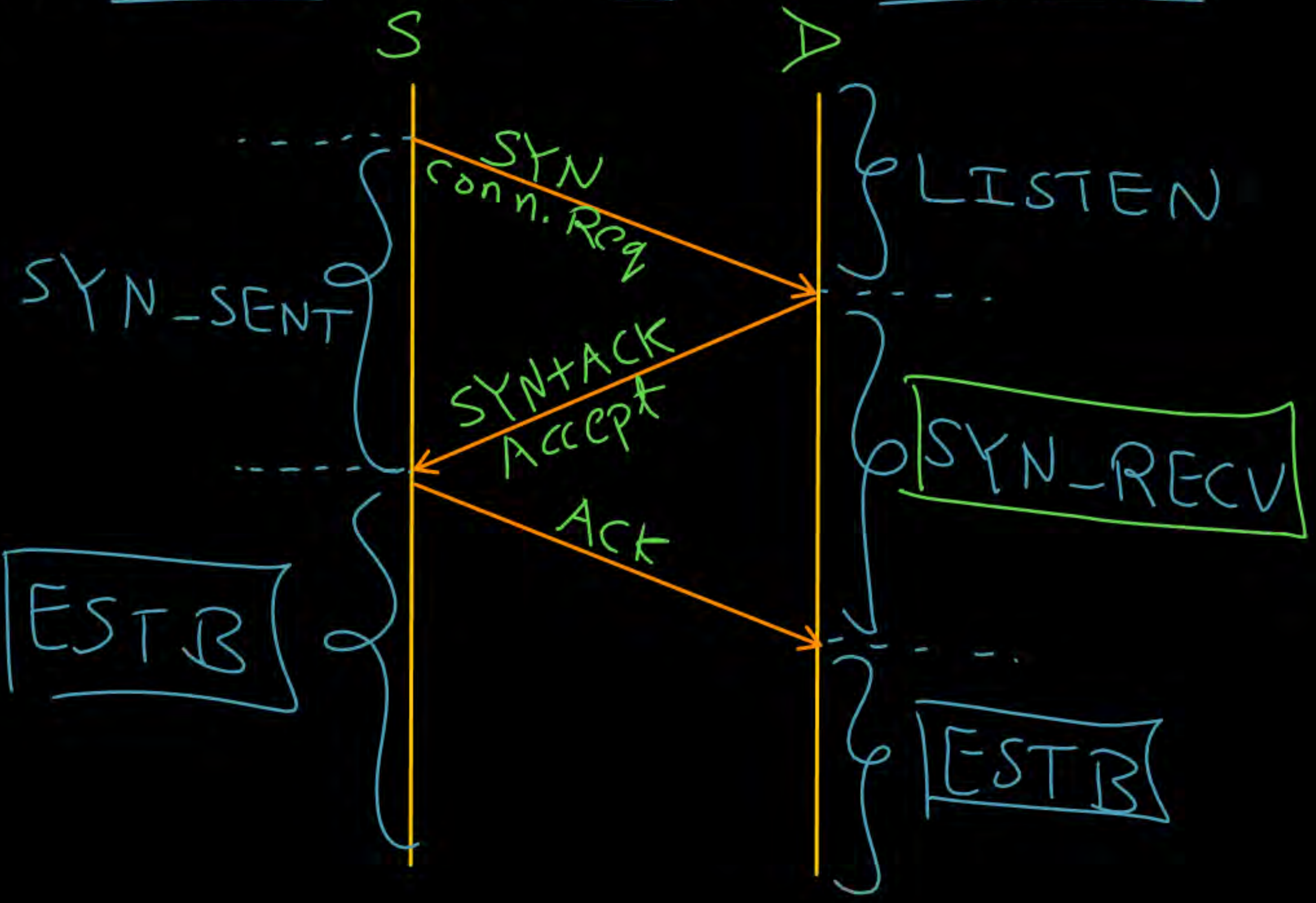
#Q. Consider the 3-way handshake protocol for TCP connection establishment between hosts S and D. Host S sends a TCP connection request message (TCP segment) to D. D accepts the connection request and send accept message (TCP segment) to S and waiting for ACK of it from S. Identify the current TCP state of Host D ?

A. ~~LISTEN~~

B. ~~SYN_SENT~~

C. SYN_RECV

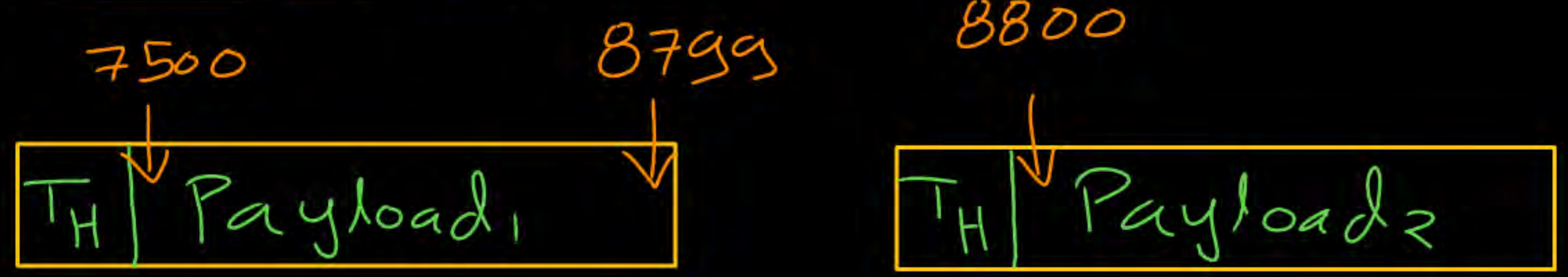
D. ~~ESTABLISHED~~



$\boxed{\text{Ans: C}}$

[NAT]

#Q. Consider a TCP sender send two segments to TCP receiver one after another. The sequence number field value in the header of the segments (in decimal) are 7500 and 8800 respectively. How many bytes carried by the first segment in its payload field ?



Seq No. = 7500

Seq. No. = 8800

$$\text{Payload}_1 \text{ Size} = (8800 - 7500) \text{ bytes}$$
$$= 1300 \text{ bytes}$$

$\boxed{\text{Ans: } 1300}$

[NAT]

#Q. Consider TCP protocol runs over a 300 Mbps network. What should be maximum segment lifetime (in seconds, rounded off to nearest integer) ?

$$MSL = \frac{2^{32}\text{ bytes}}{\text{Bandwidth}} = \frac{2^{32} * 2^{3}\text{ bits}}{300\text{ Mbps}} = \frac{2^{35}\text{ bits}}{3 * 10^{8}\text{ bits/sec}}$$

$$= 114.53\text{ sec}$$

$$\boxed{\text{Ans: 114 to 115}}$$

[NAT]

#Q. Consider new reliable byte-stream transport protocol like TCP, runs over a 200 Mbps network and maximum segment lifetime is 600 seconds. Calculate minimum number of bits required for sequence number field in the header ?

Max^m no. of bytes can be transmitted in one MSL

$= MSL * Bandwidth = 600 sec * 200 Mbps$

$= 12 * 10^{10}$ bits $= \frac{12 * 10^{10}}{8}$ bytes

Min^m no. of bits for seq. no. field

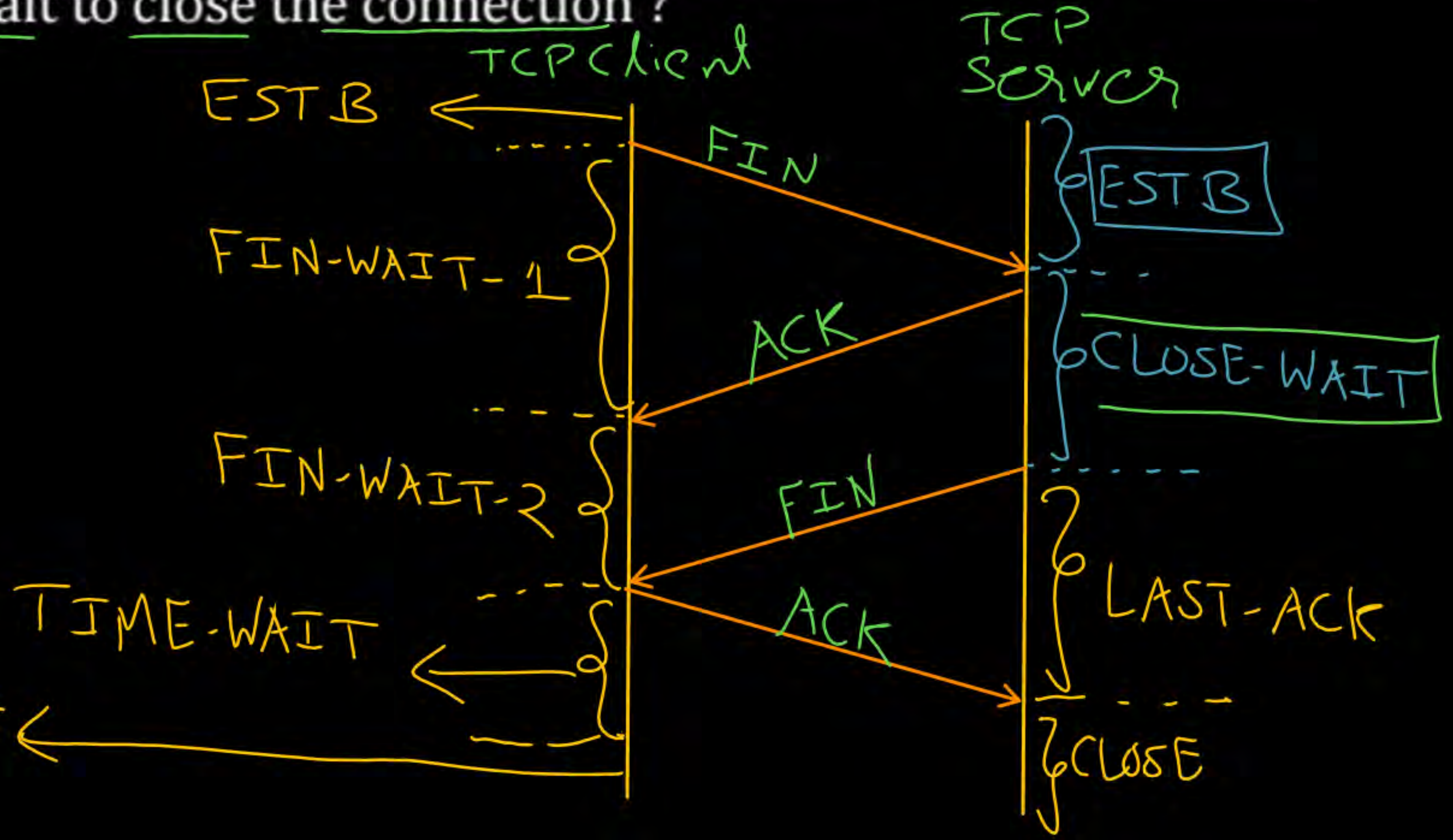
$= \left\lceil \log_2\left(\frac{12*10^{10}}{8}\right) \right\rceil$ bits $= \lceil 33.80 \rceil$ bits $= 34$ bits

$\boxed{Ans = 34}$

[MCQ]

#Q. Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. The TCP server send ACK to acknowledges the FIN segment, in which state does the server side TCP connection wait to close the connection ?

- **A** CLOSE-WAIT
- **B** TIME-WAIT
- **C** FIN-WAIT-1
- **D** FIN-WAIT-2



Ans: A

[MSQ]

#Q. Which of the following socket API function is/are executed by TCP Server ?

A. bind()

B. ~~connect()~~ → TCP client

C. listen()

D. accept()

$\boxed{\text{Ans: A, C \& D}}$

[NAT]

#Q. Consider size of congestion window of a TCP connection be 16 MSS when a timeout occurs. The round trip time of the connection is 3 seconds. The time taken (in sec) by the TCP connection to reach up to 12 MSS congestion window is ______ .

CWND = 16 MSS

Time-out

SSThresh $= \frac{CWND}{2} = \frac{16\ MSS}{2} = 8\ MSS$

RTT = 3 sec

CWND = 1 MSS $\xrightarrow{1}$ 2 MSS $\xrightarrow{2}$ 4 MSS $\xrightarrow{3}$ 8 MSS $\xrightarrow{4}$ 9 MSS $\xrightarrow{5}$ 10 MSS $\xrightarrow{6}$ 11 MSS $\xrightarrow{7}$ 12 MSS

Time $= 7 * RTT$

$= 7 * 3$ sec

$= 21$ sec

Ans: $\boxed{21 \text{ to } 24}$

[NAT]

#Q. Consider in TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm, where the window size at the start of the slow start phase is 1 MSS and the threshold at the start of the first transmission is 32 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size (in MSS) at the end of the tenth transmission ?

SSThresh = 32 MSS

CWND = 1 MSS → 1

2 MSS → 2

4 MSS → 3

8 MSS → 4

16 MSS → 5

Timeout

$SSThresh = \frac{CWND}{2} = \frac{16\,MSS}{2} = 8\,MSS$

CWND = 1 MSS → 6

2 MSS → 7

4 MSS → 8

8 MSS

CWND = 8 MSS → 9

9 MSS → 10

10 MSS

$\boxed{\text{Ans: } 10}$

THANK - YOU